

Weijia Zhang

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EDUCATION

University of Illinois Urbana-Champaign (UIUC)

B.S in Computer Science and Mathematics

Graduation Date: May 2026

GPA: 3.86/4.0 (Dean's List)

RESEARCH EXPERIENCE

U Lab, University of Illinois Urbana-Champaign

Jan 2025 – Present

Research Assistant (Advisor: Prof. Jiaxuan You)

- **SeeingEye: Agentic Information Flow Unlocks Multimodal Reasoning in Text-only LLMs** *First Author & Project Lead* Jan 2025 – Present
 - * Designed an agentic information flow that utilized a small VLM ReAct agent to “translate” VQA-tasks into structured intermediate representations (SIR) for text-only LLMs, enabling multimodal reasoning that beat larger end-to-end VLMs.
 - * Developed the SIR schema & prompting protocol and built SmartGridCaption—a visual chain-of-thought / iterative caption-and-patch module—to deliver high-fidelity, task-specific evidence to the reasoner.
 - * Implemented auditable tool selection with capability mask gates (text/table/region) and loop termination logic (confidence thresholds, evidence coverage) for efficient, reliable inference.
- **OpenManus-RL: Better Memory Planning Agent in General Environments via Reinforcement Learning** *Researcher* Jan 2025 – July 2025
 - * Implemented REACT reasoning architecture (*think*→*act*→*tool-use*) and multi-agent flow for action optimization.
 - * Built an MCP-compatible tool framework supporting automated tool invocation and output processing.
 - * Constructed SFT datasets compatible with the rigorous AgentBench environment (Gaia, WebArena, WebShop, AlfWorld).

Microsoft

July 2025 – Sep 2025

Research Intern, MSRA, DKI

- Research on LLM Agent/Multi-modal for Microsoft Copilot in Excel
- Built **TextAnalysisSFT**, a multi-threaded pipeline that generates SFT training data for the new TextAnalysis API in Excel Copilot: Kaggle mining (2000+ real world example csv)→ heavy-text filtering → query & Office.js code gen → SheetEngine validation—delivering production-quality datasets and improving Office Script code-gen accuracy 75%

University of California, Santa Barbara

Sep 2023 – May 2024

Research Assistant (Advisor: Prof. Misha Sra)

- Designed multi-agent social simulation framework to model social contract formation
- Implemented persona-encoding via prompt engineering and personalized QA sets for differentiated agent personalities
- Developed agent economic system and AI policy generation framework

PUBLICATIONS

Weijia Zhang*, Zijia Liu*, Haoru Li*, Haoqi Chen*, Jiaxuan You. “SeeingEye: Agentic Information Flow Unlocks Multimodal Reasoning in Text-only LLMs.” *arXiv preprint arXiv:2510.25092*

Gordon Dai*, **Weijia Zhang***, Devang Acharya*, Jinhan Li*, Qianyu Zhang, Victor (Qibin) Huang, Ryan Zhou, Keyi Wang, Dennis Niu, Misha Sra. “On the Democratic Society of LLM Agents.” *PersonaNLP Poster, First Workshop on LLM Persona Modeling (PersonaNLP), NeurIPS 2025 Workshop*. Mexico City.

Miaomiao Li, Hao Chen, Yang Wang, Tingyuan Zhu, **Weijia Zhang**, Kaijie Zhu, Kam-Fai Wong, Jindong Wang. “Understanding and Mitigating the Bias Inheritance in LLM-based Data Augmentation on Downstream Tasks.” *arXiv preprint arXiv:2502.04419*, under review at ACL ARR 2025.

Kunlun Zhu, Zijia Liu, Bingxuan Li, Muxin Tian, Yingxuan Yang, Jiaxun Zhang, Pengrui Han, Qipeng Xie, Fuyang Cui, **Weijia Zhang**, Xiaoteng Ma, Xiaodong Yu, Gowtham Ramesh, Jialian Wu, Zicheng Liu, Pan Lu, James Zou, Jiaxuan You. “Where LLM Agents Fail and How They Can Learn From Failures.” *arXiv preprint arXiv:2509.25370*, under review at ACL 2026.

Gordon Dai*, **Weijia Zhang***, Jinhan Li, Siqi Yang, Srihas Rao, Arthur Caetano, Misha Sra. “Artificial Leviathan: Exploring Social Evolution of LLM Agents Through the Lens of Hobbesian Social Contract Theory.” *The First Workshop on AI Behavioral Science, ACM SIGKDD 2024*

TEACHING

Course Assistant, CS 233: Computer Architecture

University of Illinois Urbana-Champaign

- Held office hours and one-on-one help sessions, answering student questions on pipelining, caching, memory hierarchy, etc.
- Co-created homework puzzles and helped build and debug the Spimbot MIPS assembly competition for students.

Exec & Workshop Lead, UIUC ACM Gamebuilders

University of Illinois Urbana-Champaign

- Led weekly workshops on game development topics, including computer graphics, Blender, Unity, C#, game AI, and performance optimization.
- Provided one-on-one technical mentoring to members on designing, implementing, and debugging their game projects.

WORK EXPERIENCE

The stu/dio at UIUC <i>Programmer (Advisor: Prof. Dan Cermak)</i> - Developed games in Unreal, Unity, and Godot, including <i>VRchaeology</i>, <i>Resilient Community</i>, and <i>RPG</i>. - Researched and implemented VR interaction mechanics and performance optimizations.	Mar 2024 – May 2025
Tencent <i>Software Engineer, Wechat Group</i> - Developed a cross-platform (Android & iOS) hardware profiler that reports performance data for Wechat Mini Program, support 200+ partner teams to surface performance bottlenecks - Built a Unity Mono Memory Profiler that found 40 + hidden allocations per pilot app and cut leak-related crashes by 120%	Aug 2024 – Sep 2024
NetEase Game <i>Software Engineer</i> - Developed a SQLite + Flask + Vue asset DB that indexes multi-terabyte game art assets in < 200 ms - Implemented a multi-threaded, Redis-cached auto-scanner that accelerates folder-to-DB ingest by 300 % - Rendered interactive mesh-to-level dependency graphs; tech artists now pinpoint broken links in < 15 min instead of 4h	July 2024 – Aug 2024
Reborn Network <i>LLM Software Engineer</i> - Developed a multiplayer Role-playing agents Unity VR Game enabling text / voice / embodied VR actions (e.g., holding hands, dancing) with real-time voice interactions (< 1s latency), boost average session length by 38%. - Introduced vector-DB long-term memory (dynamic CRUD + RAG) so agents remember players and past events, raising dialogue-coherence ratings from 2/5 to 4/5. - Created a shareable “character-card” framework, seeding a UGC ecosystem and trimming designer NPC-setup time by 300%	May 2023 – July 2023

ACTIVITES & LEADERSHIP

• UIUC ACM GameBuilder Exec/Programming Lead	Aug 2023 – Present
• RecSys 2025, Reviewer	Aug. 2025
• Global Game Jam 2023 Ranking 9th	Feb. 2023
• Ludum Dare 2023 Ranking top 10%	May. 2023
• UIUC MechMania28 AI Hackathon 2nd Place Winner	Sep. 2022